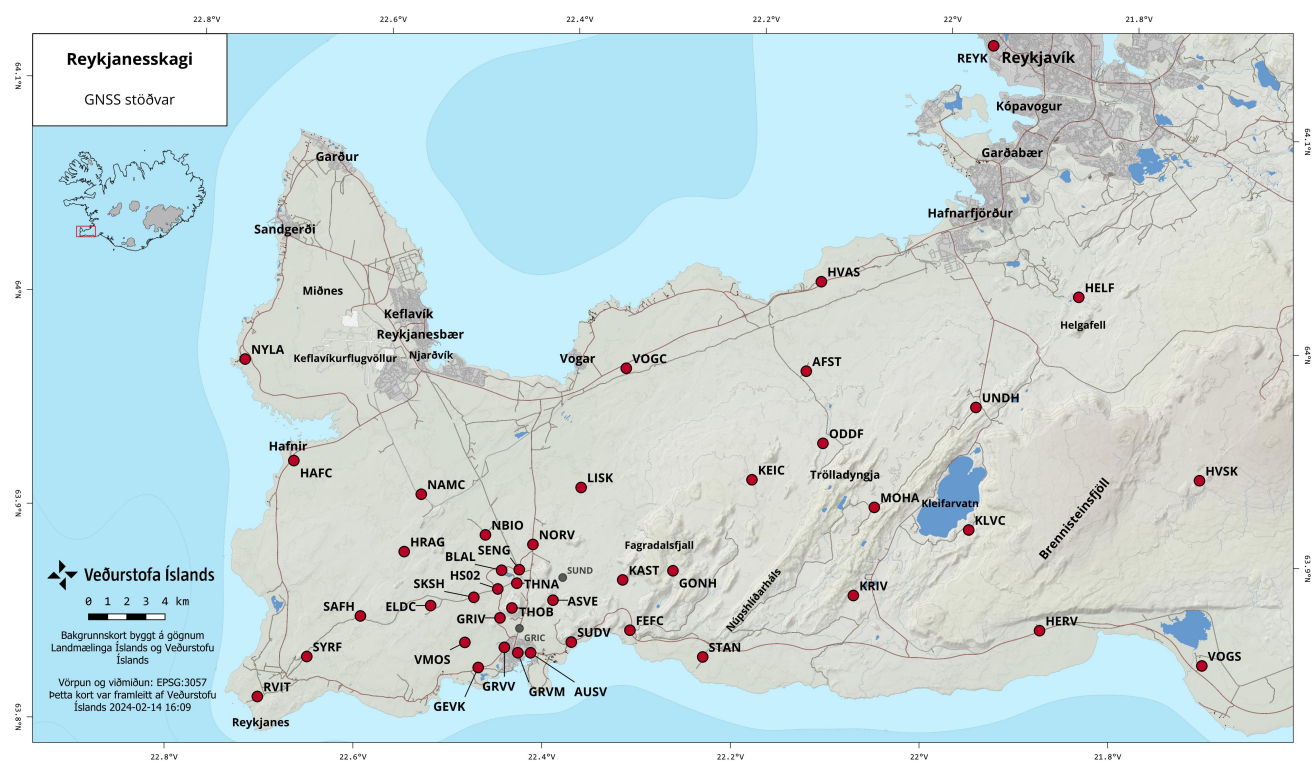


```
In [1]: import numpy as np
import pandas as pd
import plotly.express as px
import plotly.graph_objects as go
import matplotlib.pyplot as plt
#import datetime
```



<http://brunnur.vedur.is/gps/reykjanes.html>

```
In [2]: #HS02=pd.read_csv('HS02-plate.NEU.txt', sep=',')
HS02=pd.read_csv('HS02-plate.NEU-2025.txt', sep=',')
HS02
```

	date	time	dN[mm]	DN[mm]	dE[mm]	DE[mm]	dU[mm]	DU[mm]
0	2023/11/09	12:00:00	-195.29	2.61	513.52	1.88	347.15	11.61
1	2023/11/10	12:00:00	-210.32	6.32	467.17	10.60	359.11	25.30
2	2023/11/11	12:00:00	39.30	2.27	-143.06	2.18	-125.51	9.13
3	2023/11/12	12:00:00	74.57	1.69	-182.57	1.45	-126.55	7.07
4	2023/11/13	12:00:00	90.98	1.96	-210.92	1.68	-104.68	8.41
...	...	...	...	...	...	...	...	...
492	2025/04/15	12:00:00	689.69	2.05	-1414.10	1.71	604.37	8.22
493	2025/04/16	12:00:00	690.17	2.12	-1412.53	1.67	603.47	8.28
494	2025/04/18	12:00:00	690.42	2.11	-1412.77	1.70	612.14	8.70
495	2025/04/19	12:00:00	690.19	2.25	-1415.11	1.93	612.09	8.96
496	2025/04/20	12:00:00	689.39	1.97	-1415.99	1.64	615.86	8.03

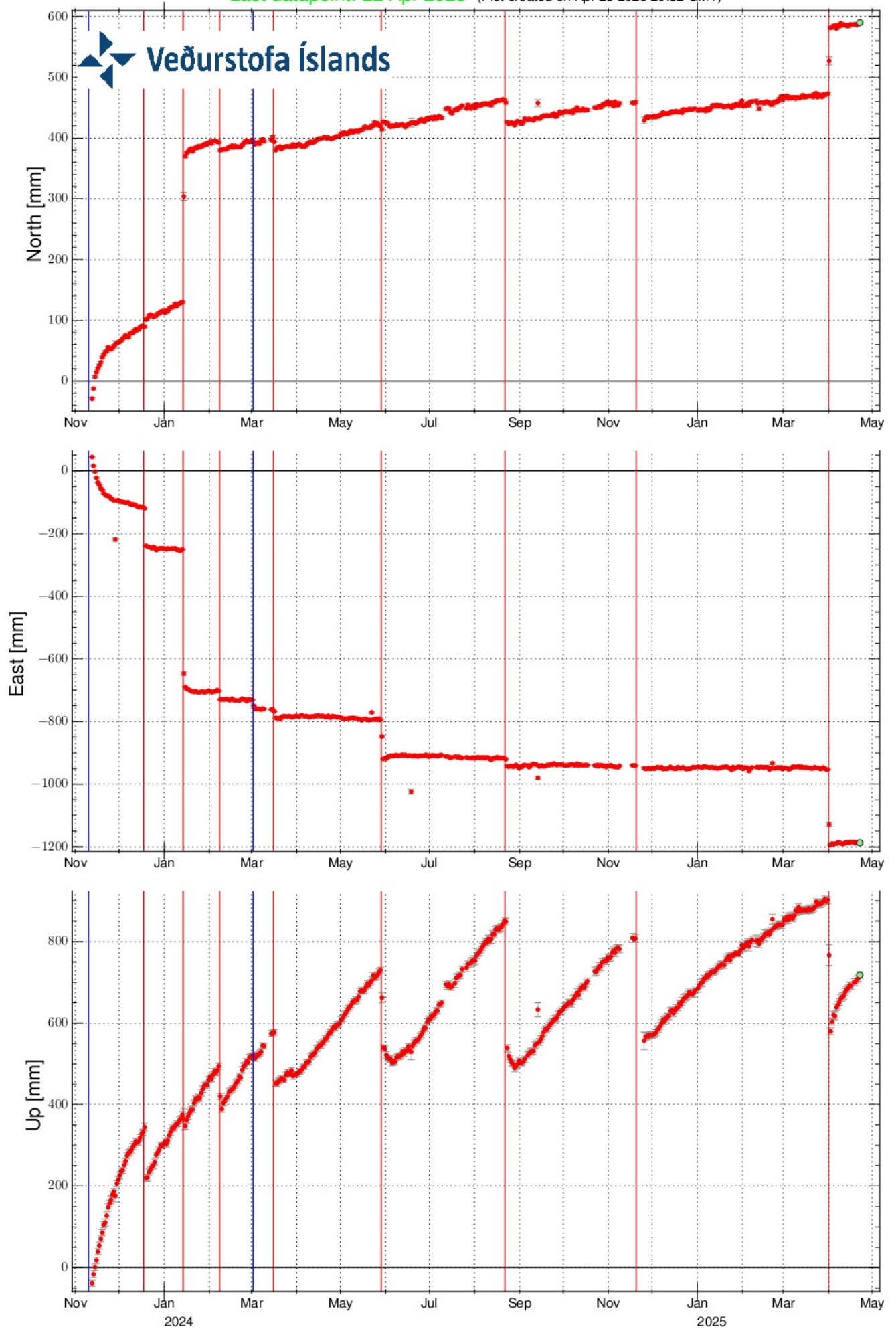
497 rows × 8 columns

In GNSS (Global Navigation Satellite System) data, the columns typically represent the displacement or movement of a GNSS station in three dimensions, relative to a reference point or initial position. The labels you've provided are likely abbreviations for specific displacement measurements:

- 1. **dN[mm]**: Change in the **north** direction, measured in millimeters. This indicates how much the station has moved north or south over time.
- 2. **dE[mm]**: Change in the **east** direction, measured in millimeters. It shows how much the station has shifted east or west.
- 3. **dU[mm]**: Change in the **up** (vertical) direction, measured in millimeters. This tracks the vertical movement, indicating whether the station has been uplifted or subsided.

HS02 (HS02) Reference frame: North American plate

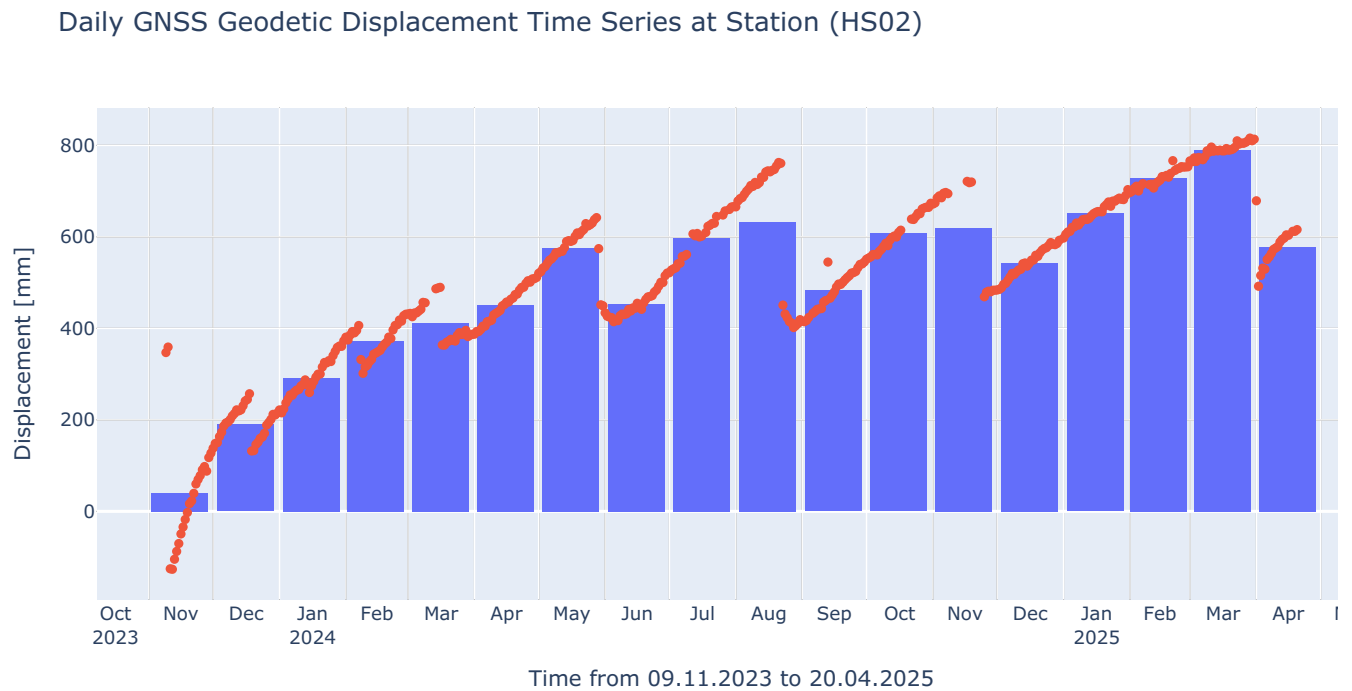
Last datapoint: 22 Apr 2025 (Plot created on Apr 23 2025 20:32 GMT)



```
In [3]: #convert the date "object" to a datetime64[ns]
date_HS02 = pd.to_datetime(HS02['date'])
dU_HS02 = HS02['dU[mm]']
```

```
#print(date_THNA,dU_THNA)
```

```
In [4]: fig = px.histogram(x=date_HS02 , y=dU_HS02, histfunc="avg",
                           title="Daily GNSS Geodetic Displacement Time Series at Station (HS02)")
fig.update_traces(xbins_size="M1")
fig.update_xaxes(showgrid=True, ticklabelmode="period", dtick="M1", tickformat="%b\n%Y")
#fig.update_layout(bargap=0.1)
fig.add_trace(go.Scatter(mode="markers", x=date_HS02, y=dU_HS02, name="Displacement [mm]"))
#fig.show()
# Set the axis titles using the labels parameter
fig.update_layout(
    bargap=0.1,
    # title='Annual Values by Category',
    xaxis_title='Time from 09.11.2023 to 20.04.2025',
    yaxis_title='Displacement [mm]',
    width=1100, # Set the width of the figure
    height=500 # Set the height of the figure
)
```

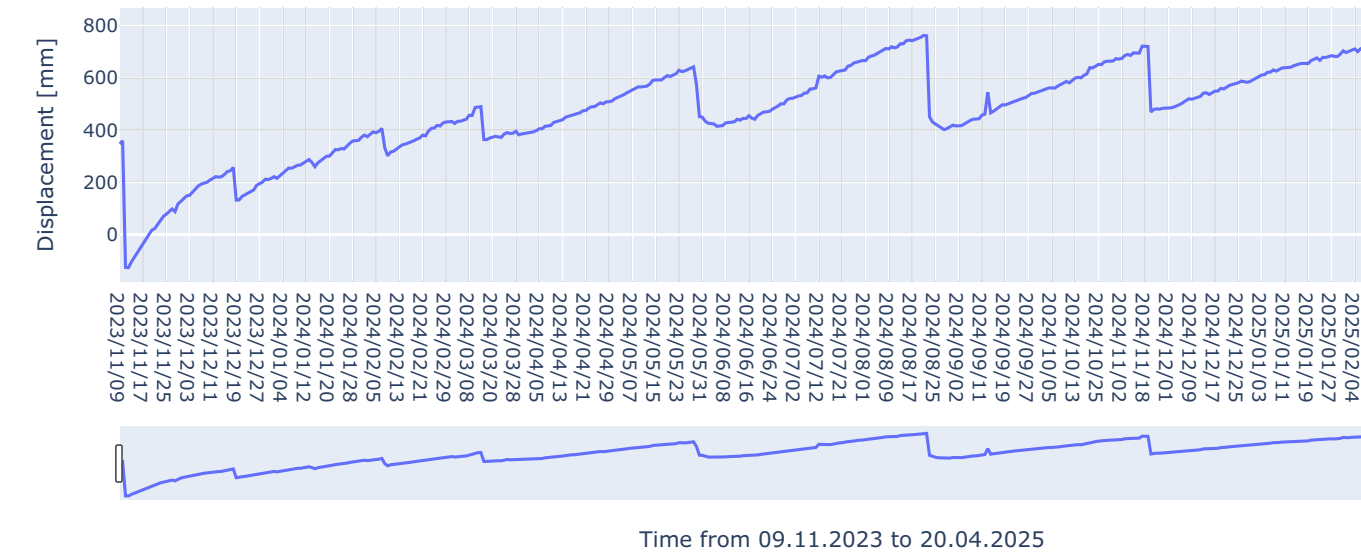


```
In [5]: fig = px.line(HS02,x='date', y='dU[mm]', range_x=['2023-11-09','2025-04-20'],
                      title='Daily GNSS Geodetic Displacement Time Series at Station (HS02)')

fig.update_xaxes(rangeslider_visible=True)

# Set the axis titles using the labels parameter
fig.update_layout(
    # title='Annual Values by Category',
    xaxis_title='Time from 09.11.2023 to 20.04.2025',
    yaxis_title='Displacement [mm]',
    width=1100, # Set the width of the figure
    height=500 # Set the height of the figure
)
```

Daily GNSS Geodetic Displacement Time Series at Station (HS02)



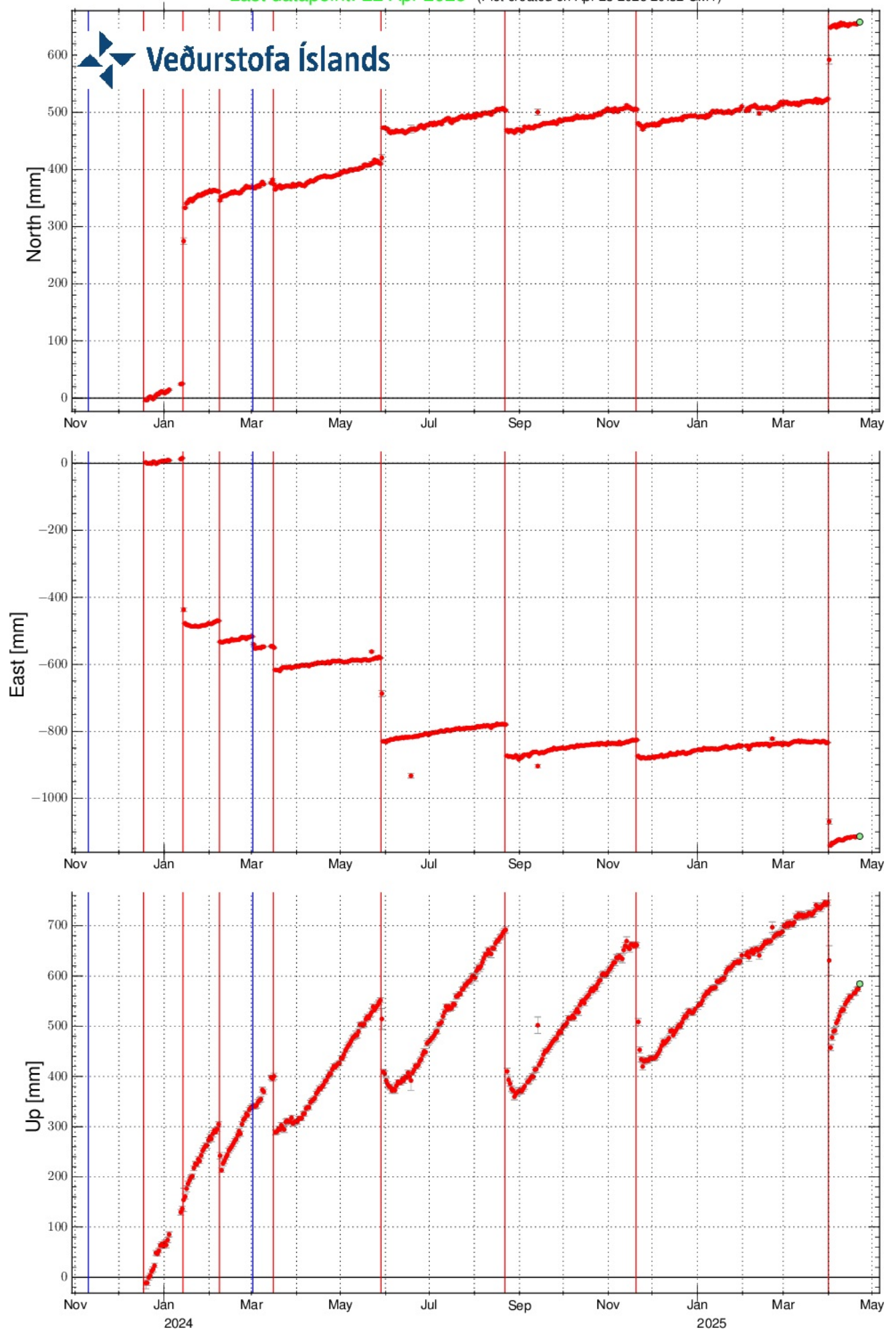
Time from 09.11.2023 to 20.04.2025

```
In [6]: THNA = pd.read_csv('THNA-plate.NEU-2025.txt', sep=',')
THNA
```

Out[6]:

	date	time	dN[mm]	DN[mm]	dE[mm]	DE[mm]	dU[mm]	DU[mm]
0	2023/12/19	12:00:00	-3.17	2.55	2.73	1.91	-11.39	11.54
1	2023/12/20	12:00:00	-3.16	1.26	-0.25	1.12	-11.12	5.08
2	2023/12/21	12:00:00	0.66	1.19	-0.28	1.05	-0.73	4.81
3	2023/12/22	12:00:00	2.57	1.14	0.53	1.03	3.12	4.56
4	2023/12/23	12:00:00	0.56	1.10	-1.22	1.01	11.75	4.45
...	...	...	...	...	...	...	...	...
468	2025/04/15	12:00:00	654.34	1.32	-1117.73	1.16	558.82	5.07
469	2025/04/16	12:00:00	654.10	1.48	-1115.69	1.29	558.96	5.61
470	2025/04/18	12:00:00	654.75	1.31	-1114.48	1.13	565.69	5.13
471	2025/04/19	12:00:00	654.76	1.46	-1115.89	1.36	566.74	5.48
472	2025/04/20	12:00:00	654.03	1.19	-1113.99	1.02	573.49	4.60

473 rows × 8 columns



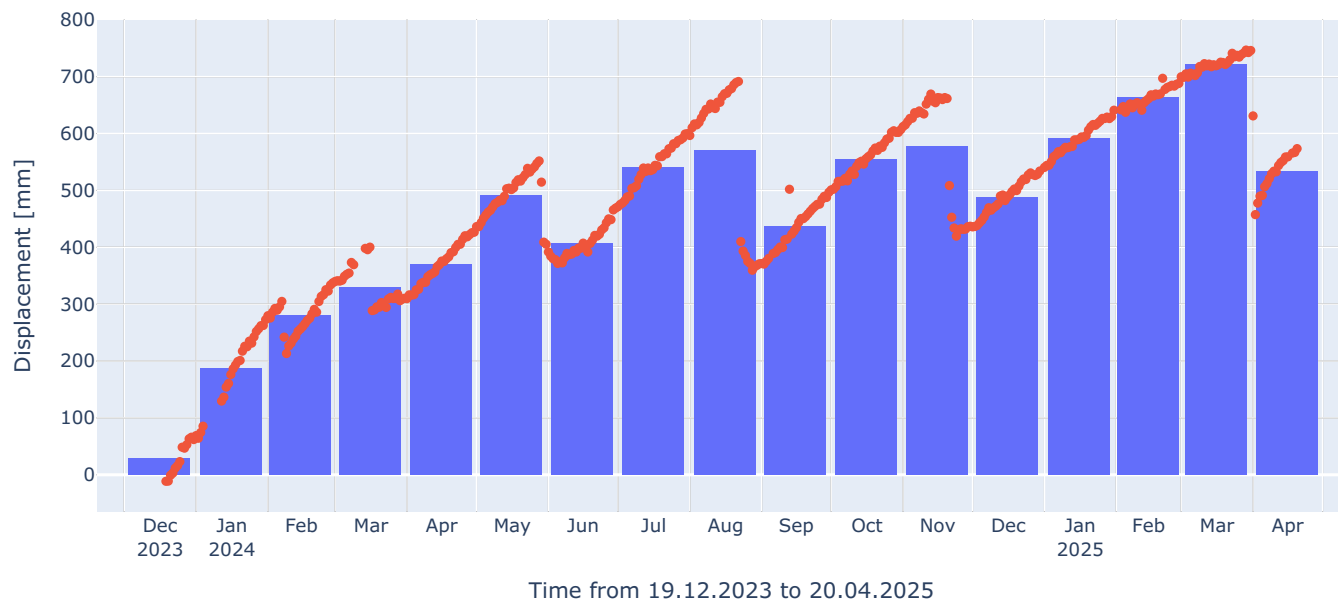
```
In [7]: #convert the date "object" to a datetime64[ns]
date_THNA = pd.to_datetime(THNA['date'])
du_THNA = THNA['dU[mm]']
```

```
#print(date_THNA,dU_THNA)
```

```
In [8]: fig = px.histogram(x=date_THNA , y=dU_THNA , histfunc="avg",
                           title="Daily GNSS Geodetic Displacement Time Series at Station ()")
fig.update_traces(xbins_size="M1")
fig.update_xaxes(showgrid=True, ticklabelmode="period", dtick="M1", tickformat="%b\n%Y")
#fig.update_layout(bargap=0.1)
fig.add_trace(go.Scatter(mode="markers", x=date_THNA, y=dU_THNA , name="Displacement [mm]"))
#fig.show()
# Set the axis titles using the labels parameter
fig.update_layout(
    bargap=0.1,
    # title='Annual Values by Category',
    xaxis_title='Time from 19.12.2023 to 20.04.2025',
    yaxis_title='Displacement [mm]',
    width=1100, # Set the width of the figure
    height=500 # Set the height of the figure
)
```



Daily GNSS Geodetic Displacement Time Series at Station ()

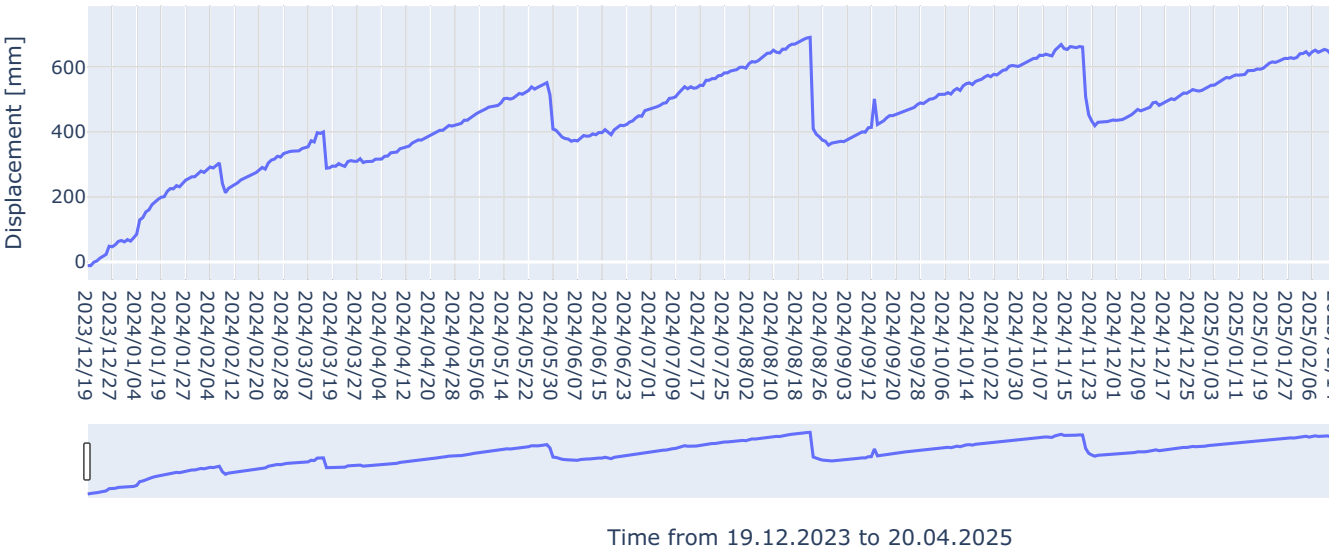


```
In [9]: fig = px.line(THNA,x='date', y='dU[mm]', range_x=['2023-11-09','2025-04-20'],
                      title='Daily GNSS Geodetic Displacement Time Series at Station (THNA)')

fig.update_xaxes(rangeslider_visible=True)

# Set the axis titles using the labels parameter
fig.update_layout(
    # title='Annual Values by Category',
    xaxis_title='Time from 19.12.2023 to 20.04.2025',
    yaxis_title='Displacement [mm]',
    width=1100, # Set the width of the figure
    height=500 # Set the height of the figure
)
```

Daily GNSS Geodetic Displacement Time Series at Station (THNA)



In []:

Loading [MathJax]/jax/output/CommonHTML/fonts/TeX/fontdata.js